

Exploratory Data Analysis on a Used Cars Dataset

Cars EDA Student Project

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Dataset Source: uncleaned_used_cars_eda.csv (mentor-provided dataset), 123 used-car listings scraped/compiled with brand, price, mileage, fuel type, transmission, ownership and location details.

1. Objective

The purpose of this project is to perform a complete Exploratory Data Analysis (EDA) on a used-car listings dataset: to understand its structure, improve its quality through systematic cleaning, and identify patterns, relationships and outliers using visualisations. No predictive model is built; the goal is understanding, not prediction.

The main questions explored are:

- Which brands and fuel types are most common among the listings?
- How are used-car prices distributed?
- Does vehicle age relate to price, as commonly expected with depreciation?
- Does kilometres driven appear to influence price?
- Do automatic and manual cars differ meaningfully in price?
- Which numerical features are most strongly correlated with price?
- Which brands have the highest and lowest average prices?
- Where are the important outliers or unusual records in this data?

2. Dataset Overview

The raw dataset contains **123 rows and 15 columns**, covering used-car listings across 8 brands and 6 Indian cities. Each row represents one car listing, with attributes for brand, model, year, price, kilometres driven, fuel type, transmission, ownership history, mileage, engine size, city, insurance status, seller rating and listing date.

Column	Description	Data Type
car_id	Unique listing identifier	Text
brand / model	Car manufacturer and model name	Text
year	Manufacturing year	Text (needed conversion)
price_inr	Listed price in INR	Numeric
km_driven	Kilometres driven	Text (needed conversion)
fuel_type	Petrol / Diesel / CNG / Electric	Text (category)
transmission	Manual / Automatic	Text (category)
owner_type	Ownership history (1st/2nd/3rd)	Text (category)
mileage_kmpl	Fuel efficiency (km per litre)	Numeric
engine_cc	Engine displacement (cc)	Numeric
city	Listing city	Text (category)
insurance_valid	Whether insurance is valid	Text (Yes/No)
seller_rating	Seller rating (expected 1-5)	Numeric
listed_date	Date the car was listed	Text (mixed formats)

Sample records and full column-by-column structure were inspected using `df.shape`, `df.dtypes` and `df.head()` in the accompanying code file.

3. Data Quality and Cleaning

The raw file contained several realistic data-quality problems that needed to be resolved before analysis:

- **3 fully duplicated rows** (identical car_id and all fields) — dropped, since they would double-count those listings.
- **Inconsistent text casing/whitespace** in brand, fuel_type, transmission, owner_type, city and insurance_valid (e.g. 'MARUTI' vs 'maruti', 'diesel ' with a trailing space) — standardised with trimming and title-casing.
- **A brand typo** ('Toyta') — corrected to 'Toyota'.
- **City name aliases** for the same city (Hyd/Hyderabad, Bangalore/Bengaluru, Delhi/New Delhi) — merged into one label each.
- **'year' stored as text** with a placeholder value 'unknown' in 4 rows — converted to numeric, with 'unknown' treated as missing rather than guessed.
- **'km_driven' stored as text**, with a ' km' unit suffix in some rows and one negative value (-1200, impossible for a distance) — cleaned to numeric, negative value set to missing.
- **'seller_rating' out-of-range values**: 5 rows showed 8.7 on a scale that should run 1-5 — treated as a data-entry error and set to missing.
- **'listed_date' mixed formats** (DD/MM/YYYY and YYYY-MM-DD) with at least one impossible date (31 February) — parsed per format, invalid dates set to missing.
- **Missing values** in price_inr, mileage_kmpl and engine_cc (each under 5% missing) — imputed with the column median, which is robust to the outlier already present in price.
- **Missing brand/transmission** values with no reliable way to infer them — labelled explicitly as 'Unknown' rather than dropped or guessed.

Dataset size before and after cleaning:

Stage	Rows	Columns
Before cleaning	123	15
After cleaning	120	17 (2 derived columns added)

Two columns were added during cleaning: **car_age** (2026 minus year, more directly interpretable than raw year) and **is_price_outlier** (a flag for the one extreme price, used to exclude it from relationship charts without deleting it from the dataset).

4. Descriptive Analysis

Summary statistics for the numerical columns (after cleaning):

Metric	Price (INR)	Km Driven	Mileage (kmpl)	Engine (cc)	Rating	Age (yrs)
Mean	7,98,475	76,863	16.1	1,503	4.0	10.1
Median	7,24,000	73,500	16.5	1,500	4.0	10.0
Std Dev	8,59,549	64,896	2.6	452	0.6	5.0
Min	1,89,000	9,000	12.0	800	3.0	2.0
Max	95,00,000	6,50,000	20.1	2,200	5.0	18.0

The mean price (7,98,475) sits well above the median (7,24,000), confirming a right-skewed distribution driven by the small number of high-priced listings, including the outlier discussed below. The km_driven range (9,000 to 6,50,000) and wide standard deviation shows listings span both nearly-new and heavily-used cars.

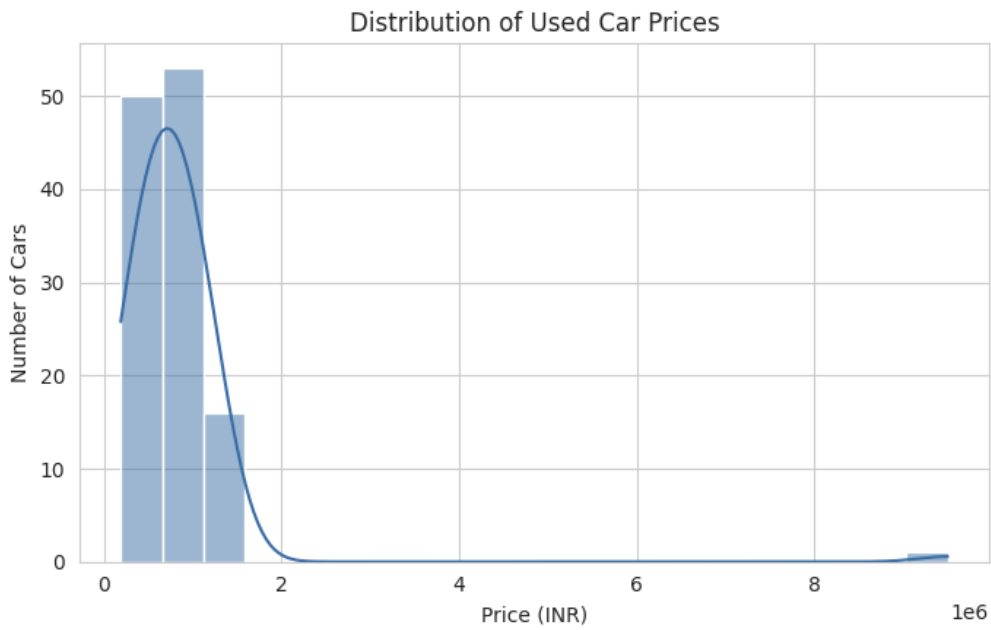
Key categorical breakdowns:

Category	Top values (count)
Brand	Maruti (29), Hyundai (19), Toyota (19), Tata (18)
Fuel Type	Petrol (52), Diesel (33), CNG (17), Electric (17)
Transmission	Automatic (69), Manual (47), Unknown (3)
Owner Type	First (40), Second (40), Third (40)
City	Hyderabad (36), Bengaluru (24), Delhi (24), Mumbai/Pune/Chennai (12 each)
Insurance Valid	Yes (93), No (27)

5. Visual EDA

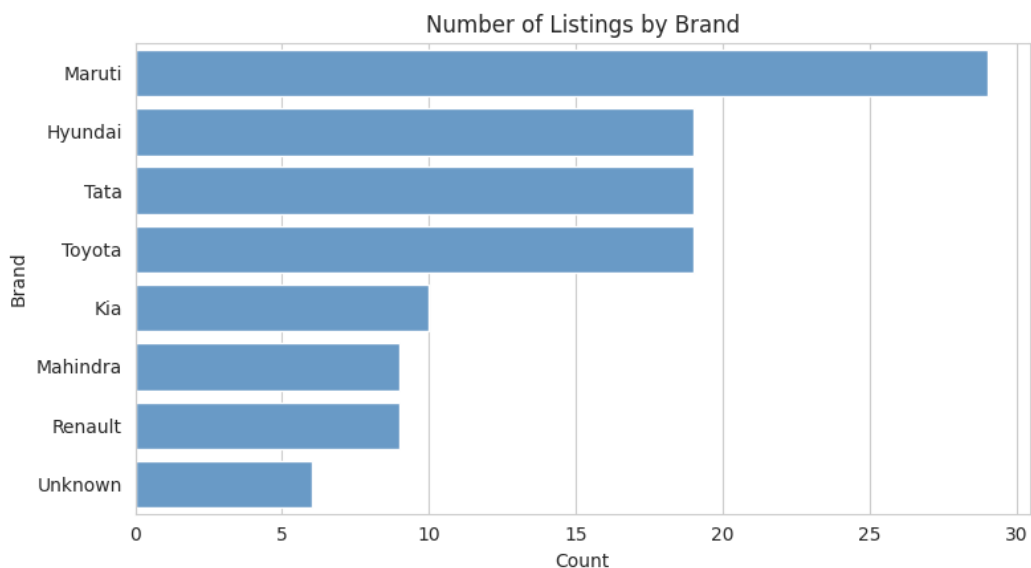
Eight charts were produced across four chart types (histogram, count plot, scatter plot, box plot, bar chart, heatmap), each with a written observation below it. The one extreme price outlier is excluded from the relationship charts (age, km, transmission, correlation) to avoid distorting the scale, but is retained in the dataset itself.

Chart 1: Distribution of Used Car Prices (Histogram)



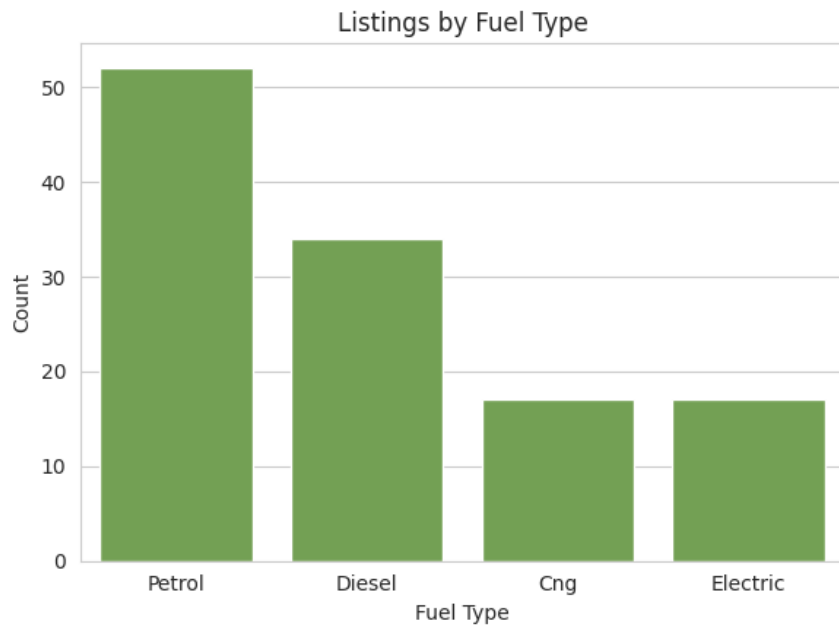
Observation: Prices are right-skewed; most cars sit under 13 lakh, with a long tail caused by a few premium/outlier listings.

Chart 2: Number of Listings by Brand (Count Plot)



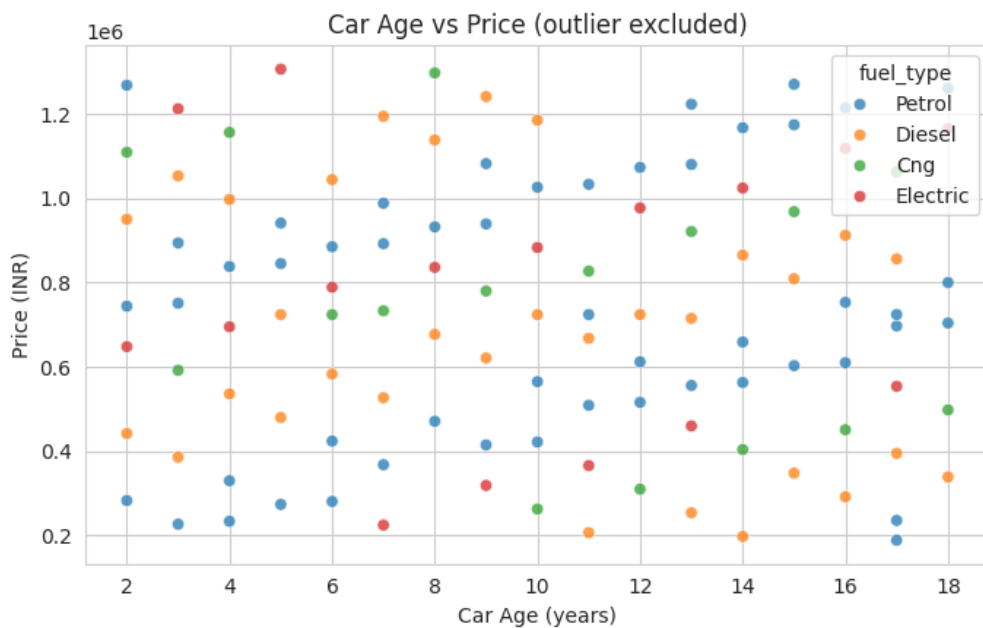
Observation: Maruti, Hyundai and Tata dominate the listings, consistent with their large share of the used-car market in India.

Chart 3: Listings by Fuel Type (Count Plot)



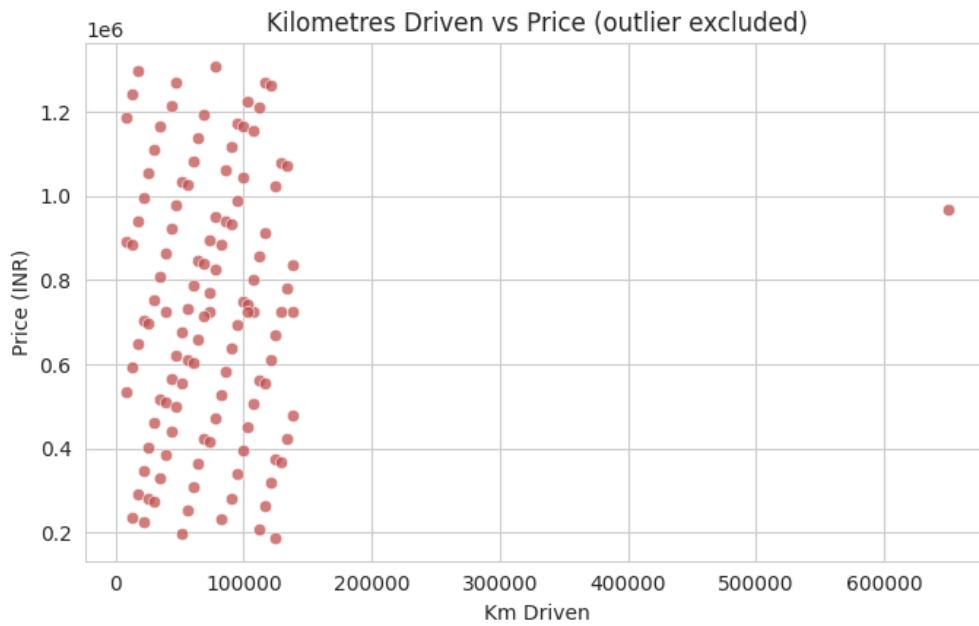
Observation: Petrol and diesel cars make up the large majority of listings; CNG and electric cars are a small minority.

Chart 4: Car Age vs Price (Scatter Plot)



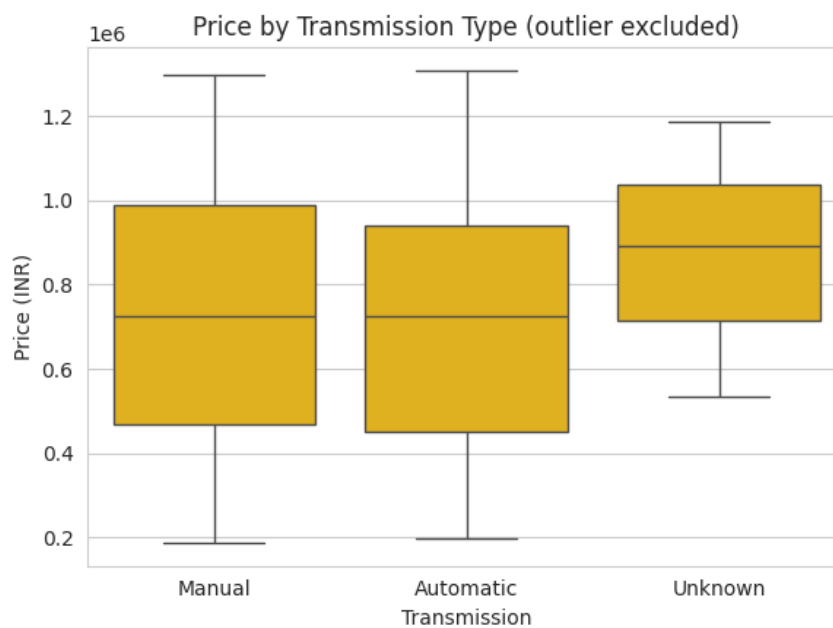
Observation: Points are scattered with no visible downward trend; car age does not appear to drive price in this dataset, unlike the depreciation pattern usually expected in used-car data.

Chart 5: Kilometres Driven vs Price (Scatter Plot)



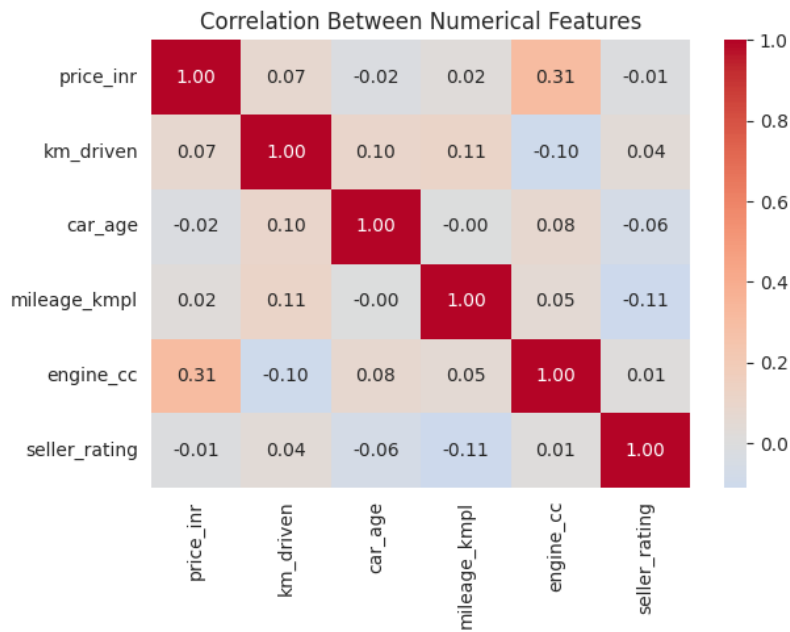
Observation: Similar to car age, kilometres driven shows no clear linear relationship with price here -- high-mileage cars are not consistently cheaper than low-mileage ones in this sample.

Chart 6: Price by Transmission Type (Box Plot)



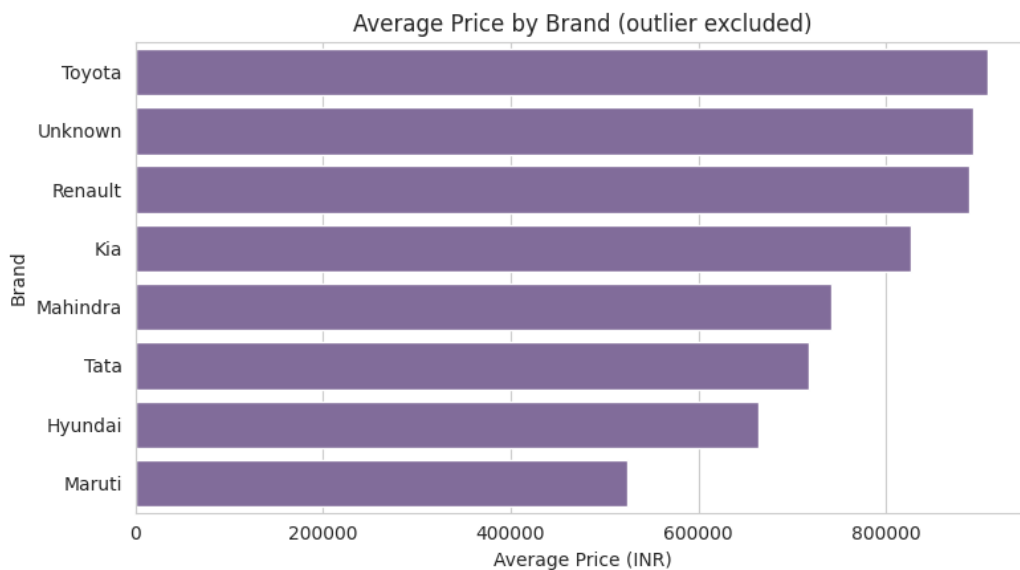
Observation: Automatic and manual cars have almost identical median prices in this dataset -- transmission type is not a strong price differentiator here.

Chart 7: Correlation Between Numerical Features (Heatmap)



Observation: Engine size has the strongest relationship with price (moderate positive correlation); age, km driven, mileage and seller rating are all weakly correlated with price, close to zero.

Chart 8: Average Price by Brand (Bar Chart)



Observation: Toyota and Kia have the highest average listed prices among the brands present in this dataset, while Maruti has the lowest despite being the most frequently listed.

6. Key Findings

- **Engine size is the strongest numeric driver of price** in this dataset (moderate positive correlation, $r \approx 0.31$) -- larger-engine cars tend to be priced higher, more so than any other numeric feature.
- **Car age and kilometres driven are only weakly related to price** ($r \approx -0.02$ and 0.07 respectively), contrary to the depreciation pattern usually expected in used-car data. This is a genuine finding of this sample, not an assumption, and limits how far this dataset can be used to model pricing by age or mileage alone.
- **Transmission type makes little difference to price:** automatic (median INR 7,24,000) and manual (median INR 7,24,000) cars have almost identical median prices.
- **Popularity and price position differ by brand:** Maruti is the most frequently listed brand, while Toyota commands the highest average price and Maruti the lowest.
- **One clear price outlier exists** (~INR 95,00,000, roughly 7x the next-highest price) with no matching difference in age, mileage or engine size -- likely a data entry error or an unusually premium vehicle that should be verified before use in any pricing model.
- **The raw data required real cleaning** before it could be trusted: duplicated rows, inconsistent category labels, a text unit mixed into a numeric column, a negative distance value, and an out-of-range rating value were all identified and resolved.

Limitations

The dataset is small (123 rows before cleaning, 120 after) and covers only 8 brands and 6 Indian cities, so findings may not generalise to the broader used-car market. A small number of values in price, mileage and engine size were imputed with column medians rather than true readings. The weak relationship between age/mileage and price may be specific to this sample and should be re-checked on a larger dataset before drawing firm conclusions about used-car pricing in general.

7. Conclusion

This project involved cleaning a genuinely messy used-car dataset -- handling duplicate rows, inconsistent category labels, mixed data types, an impossible negative value, mixed date formats, and an out-of-range rating -- before it could support reliable analysis. The biggest challenge was resisting the temptation to assume the usual used-car pricing patterns (e.g. that older or higher-mileage cars are always cheaper); the actual correlations in this sample showed those relationships were much weaker than expected, while engine size turned out to be the more meaningful price driver. This reinforced the importance of checking assumptions against the data rather than reporting textbook expectations. With a larger dataset, future analysis could explore price prediction more formally, examine regional (city-level) pricing differences in more depth, or investigate why the age/mileage-price relationship is so weak in this particular sample.